Sri Chaitanya IIT Academy.,India.

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A right Choice for the Real Aspirant

ICON Central Office - Madhapur - Hyderabad

SEC: Sr.S60(N&S), Elite, IAS & LIIT | JEE-MAIN_Shift-1_Batch-II | Date: 18-01-2026

Time: 09.00Am to 12.00Pm | GTM-15 | Max. Marks: 300

KEY SHEET

MATHEMATICS

1	**1**	2	**3**	3	**1**	4	**4**	5	**1**
6	**3**	7	**4**	8	**3**	9	**4**	10	**2**
11	**4**	12	**1**	13	**2**	14	**3**	15	**1**
16	**4**	17	**3**	18	**4**	19	**3**	20	**3**
21	**98**	22	**25**	23	**297**	24	**23**	25	**3780**

PHYSICS

26	**2**	27	**1**	28	**4**	29	**2**	30	**1**
31	**3**	32	**2**	33	**2**	34	**3**	35	**1**
36	**2**	37	**4**	38	**1**	39	**2**	40	**2**
41	**1**	42	**1**	43	**1**	44	**4**	45	**1**
46	**2**	47	**1**	48	**25**	49	**3**	50	**5**

CHEMISTRY

51	**4**	52	**1**	53	**1**	54	**2**	55	**4**
56	**2**	57	**2**	58	**3**	59	**3**	60	**3**
61	**4**	62	**3**	63	**1**	64	**4**	65	**3**
66	**1**	67	**4**	68	**2**	69	**1**	70	**1**
71	**4**	72	**11**	73	**12**	74	**180**	75	**5**

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SOLUTION

MATHEMATICS

1. Statement I : $T_m = a+(m-1)d = 1/n$ and $t_n = a+(n-1)d = 1/m$

$\Rightarrow (m-n)d = 1/n - 1/m = (m-n)/mn \Rightarrow d = 1/mn$

$\therefore\ a = \dfrac{1}{mn}$

Now, $t_{mn} = a+(mn-1)d = \dfrac{1}{mn}+(mn-1)\dfrac{1}{mn} = \dfrac{1}{mn}+1-\dfrac{1}{mn} = 1$

Statement II: $t_{mn} = a+(mn-1)d = \dfrac{1}{mn}+(mn-1)\dfrac{1}{mn} = \dfrac{1}{mn}+1-\dfrac{1}{mn} = 1$

It is possible when $15a-3b=0, 3b-5c=0$ and $5c-15a=0$

$\Rightarrow 15a = 3b \Rightarrow b = 5a$

$\Rightarrow b = \dfrac{5c}{3}, a = \dfrac{c}{3} \Rightarrow a+b = \dfrac{c}{3}+\dfrac{5c}{3} = \dfrac{6c}{3}$

$\Rightarrow a+b = 2c \Rightarrow b, c, a$ are in A.P

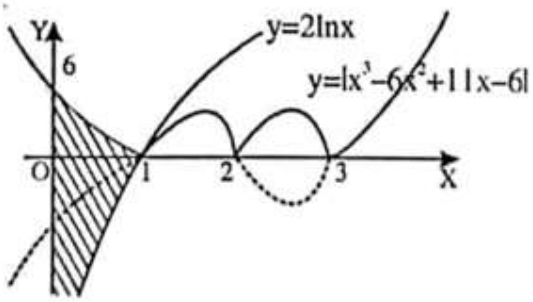
2. Number of divisors $= \dfrac{(3+1)(3+1)(9+1)}{2} - 1 = 79$

3. Take $x = y = 1 \Rightarrow f(1) = 0$

Now, put $y = \dfrac{1}{x} \Rightarrow 0 = f\left(x\dfrac{1}{x}\right) = f(x)+f\left(\dfrac{1}{x}\right) \Rightarrow f\left(\dfrac{1}{x}\right) = -f(x)$

$\therefore f\left(\dfrac{x}{y}\right) = f(x)+f\left(\dfrac{1}{y}\right) = f(x)-f(y)$

Now, $f'(x) = \lim\limits_{h\to 0}\dfrac{f(x+h)-f(x)}{h}$



$= \lim\limits_{h\to 0}\dfrac{1}{h}f\left(\dfrac{x+h}{x}\right) = \lim\limits_{h\to 0}\dfrac{f\left(1+\dfrac{h}{x}\right)-f(1)}{\dfrac{h}{x}x}$

$= \dfrac{f'(1)}{x} = \dfrac{2}{x} \Rightarrow f(x) = 2\ln x + c \Rightarrow c = 0 \Rightarrow f(x) = 2\ln x$

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The required area

$$=\int_0^1 (x^3-6x^2+11x-6)dx+\int_{-\infty}^{0} e^{y/2}dy=\frac{17}{4}$$

(Using horizontal strip for the area below the x – axis)

4. Since $x-1\le[x]\le x$ for all $x\in R$ so

$$\frac{1}{x^3}-1\le\left[\frac{1}{x^3}\right]\le\frac{1}{x^3} \qquad \Rightarrow x^8\left(\frac{1}{x^3}-1\right)\le x^5\left[\frac{1}{x^3}\right]\le x^5 \text{ for all } x$$

But $\lim\limits_{x\to0} x^5=0=\lim\limits_{x\to0}\left(x^5-x^8\right)$ So $\lim\limits_{x\to0} x^8\left[\frac{1}{x^3}\right]=0\in I\subseteq Q$

b) $\lim\limits_{x\to0}\dfrac{(1+x^3)-\left(1-\frac{x^3}{3}\right)+\left(1-\frac{x^2}{2}-1\right)x}{x^3}=\frac{5}{6}$

5.

$$f(x)=x\cos x-\sin x$$

$$f'(x)=-x\sin x+\cos x-\cos x$$

$$g(x)=x\sin x+\cos x$$

$$g'(x)=x\cos x+\sin x-\sin x$$

$$f(x)g'(x)=x^2\cos^2 x-x\cos x\sin x$$

$$g(x)f'(x)=-x^2\sin^2 x-x\sin x\cos x$$

$$\frac{f(x)g'(x)-g(x)f'(x)}{x^2}=x^2$$

$$\int\frac{f(x)g'(x)-g(x)f'(x)}{f(x)g(x)}=\log\left|\frac{g(x)}{f(x)}\right|=\log\left|\frac{x\sin x+\cos x}{x\cos x-\sin x}\right|+C$$

6. $2ae=6$ *and* $\frac{2a}{e}=12$

$$\Rightarrow ae=3 \qquad .....(i)$$

and $\frac{a}{e}=6\Rightarrow e=\frac{a}{6}$ .....(*ii*)

$\Rightarrow a^2=18$ [From (i) and (ii)]

$\Rightarrow b^2=a^2-a^2e^2=18-9=9$

$\therefore$ Latus rectum $=\frac{2b^2}{a}=\frac{2\times9}{3\sqrt{2}}=3\sqrt{2}$

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7. $f(x)=\sin x+\sin x\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} f(t)dt+\cos x\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} tf(t)dt, \quad f(x)=(A+1)\sin x+B\cos x$

$$A=\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} f(t)dt=\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}}\left((A+1)\sin t+B\cos t\right)dt \quad =2\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} B\cos t\,dt=2B \quad A=2B$$

$$B=\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} tf(t)dt=\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}}\left((A+1)t\sin t+Bt\cos t\right)dt=2(A+1)\int_0^{\frac{\pi}{2}} \underset{I}{t}\underset{II}{\sin t}\,dt$$

$$\Rightarrow B=2(A+1)\Rightarrow f(x)=-\frac{1}{3}\sin x-\frac{2}{3}\cos x, \quad f(x)\in\left[-\frac{\sqrt{5}}{3},\frac{\sqrt{5}}{3}\right]$$

8. A number will be divisible by 275, if it is divisible by 25 and 11
$x+y=1+3+4+5+6+7+8+9=43$
For divisible by 25 last two digits must be 75; (note that the digit and '0' is not there)
Hence Now we have 1, 3, 4, 6, 8, 9 to be filled in six places, such that (also x – y = 11)
sum of even places = 27 – 5 = 22 or 16 – 5 = 11
Sum of odd places = 27 – 5 = 22 or 16 – 7 = 9
2nd option is possible when we have 1, 4, 6 at even places and 3, 8, 9 at odd places.

9. Let $R=\{(3,3)(5,5),(9,9),(12,12),(5,12),(3,9),(3,12),(3,5)\}$ be a relation on the set $A=\{3,5,9,12\}$, Clearly, every element of A is related to itself. Therefore it is a reflexive.
Now, R is not symmetry because 3 is related to 5 but 5 is not related to 3
Also R is transitive relation because it satisfies the property that is a R b and b R c then a R c.

10.

$$\left|\frac{1+x^2}{2x}\right|\le\left|1+x^2\right|\le|2x|, \qquad (|x|-1)^2\le 0, only\ at\ x=\pm 1$$

$$T_n=\sin^{-1}\left[\frac{\sqrt{n}-\sqrt{n-1}}{\sqrt{n(n+1)}}\right]=\sin^{-1}\left(\frac{1}{\sqrt{n}}\left(1-\frac{1}{1+n}-\frac{1}{\sqrt{1+n}}\sqrt{1-\frac{1}{n}}\right)\right)$$

$$=\sin^{-1}\frac{1}{\sqrt{n}}-\sin^{-1}\left(\frac{1}{\sqrt{1+n}}\right), n=1,2,3 and\ add\ \therefore C=\frac{\pi}{2}$$

11. We have $7+4i=(z+\bar{w})(\bar{z}-w)=|z|^2-|w|^2-(zw-\overline{zw})$

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Since $|z|^2$ and $|w|^2$ are real, and $\left(zw-\overline{zw}\right)=0$ then

$|z|^2-|w|^2=7$ *and* $zw-\overline{zw}=-4i$

As $|z|=3.|w|^2=2.$ *Then*

$\left(z+2\overline{w}\right)\left(\overline{z}-2w\right)=|z|^2-4|w|^2-2\left(zw-\overline{zw}\right)=9-8+8i=1+8i$

12. $(1+ax)^n=1+\frac{\alpha n}{1!}x+\frac{\alpha^2n^2n(n-1)}{2!}+.....$

$=1+8x+24x^2+.....\alpha n=8$ *and* $\alpha^2n(n-1)=48\Rightarrow\alpha=2,n=4,$ *Now* $P(\alpha,n)=(2,4)$

L=Length of tangent $=\sqrt{20}$

$AB=\frac{2LR}{\sqrt{L+R}}$ R=Radius = 5 $AB=\frac{2\times2\sqrt{5}\times5}{\sqrt{45}}=\frac{20}{3}$

13. Since, f is increasing

$\therefore\quad f(x)<f(5x)<f(7x)$

$\therefore\quad \frac{f(x)}{f(x)}<\frac{f(5x)}{f(x)}<\frac{f(7x)}{f(x)}\quad\left\{\because\lim_{x\to\infty}\frac{f(7x)}{f(x)}=1\right\}$

$1<\lim_{x\to\infty}\frac{f(5x)}{f(x)}<1\qquad\lim_{x\to\infty}\left[\frac{f(5x)}{f(x)}-1\right]=0$

14. $\cos x dy-(\sin x)ydx=6xdx$

$\Rightarrow\int d(y\cos x)=\int 6xdx\Rightarrow y\cos x=3x^2+C\quad .....(i)$

Given $y\left(\frac{\pi}{3}\right)=0$

Putting $x=\frac{\pi}{3}$ and $y=0$ in eq (i) we get

$(10)\times\left(\frac{1}{2}\right)=\frac{3\pi^2}{9}+C\Rightarrow C=\frac{-\pi^2}{9}$

So, from (1) $y\cos x=3x^2-\frac{\pi^2}{2}$

Now, put $x=\frac{\pi}{6}$ in the above equation

$y\frac{\sqrt{3}}{2}=\frac{3\pi^3}{36}-\frac{\pi^2}{3}\Rightarrow\frac{\sqrt{3}y}{2}=\frac{-3\pi^2}{12}\Rightarrow y=\frac{-\pi^2}{2\sqrt{3}}$

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15.

$$x^{2026} = Q(x)(x-2)(x-3) + \underbrace{ax+b}_{P(x)}$$

Put $x = 20$ *and* $x = 3$

$2^{2026} = 2a + b.....(1) \quad 3^{2026} = 3a + b.....(2)$

$(2)-(1)\ \ a = 3^{2026} - 2^{2026}$

$b = 2^{2026} - 2a$

$b = 2^{2026} - 2.3^{2026} + 2.2^{2026}$

$= 3.2^{2026} - 2.3^{2026}$

$= 2.3\left(2^{2025} - 2.3^{2025}\right)$

$2[x] + 2 = 3[x] - 6$

$8 = [x]$

16.

Urn	Red	Black
A	4	6
B	5	5
C	λ	4

Let E be the event to set red ball from 'C'

$$P(C/E) = \frac{P(C).P(E/C)}{P(A)P(E/A) + P(B)P(E/B) + P(C)P(E/C)}$$

Given $\quad 0.4 = \dfrac{\frac{1}{3} \times \frac{\lambda}{\lambda+4}}{\frac{1}{3}\left(\frac{4}{10} + \frac{5}{10} + \frac{\lambda}{\lambda+4}\right)}$

$$\frac{4}{10}\left(\frac{9}{10} + \frac{\lambda}{\lambda+4}\right) = \frac{\lambda}{\lambda+4}$$

$76\lambda + 144 = 100\lambda \Rightarrow 24\lambda = 144 \Rightarrow y^2 = 6x$

Given parabola $y^2 = \lambda x \Rightarrow y^2 = 6x$

$\sin 30^o - \dfrac{BD}{a} \Rightarrow BD - \dfrac{a}{2}$

$\cos 30^o = \dfrac{AD}{a} \Rightarrow AD = \dfrac{\sqrt{3}}{2}a$

$B = \left(\dfrac{\sqrt{3}}{2}a, \dfrac{a}{2}\right)$ Lies on parabola $y^2 = 12x$

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17. Given vectors $\vec{a}$ and $\vec{b}$ with equal magnitudes $\dfrac{|\vec{a}+\vec{b}|+|\vec{a}-\vec{b}|}{|\vec{a}+\vec{b}|-|\vec{a}-\vec{b}|}=\sqrt{3}+2$

Apply componendo and Dividendo $\dfrac{2|\vec{a}+\vec{b}|}{2|\vec{a}-\vec{b}|}=\sqrt{3}\Rightarrow \bar{a}.\bar{b}=\dfrac{1}{2}$

18. $\dfrac{3^{200}}{8}=\dfrac{1}{8}(9^{100})$

$=\dfrac{1}{8}(1+8)^{100}=\dfrac{1}{8}\left[1+n.8+\dfrac{n(n+1)}{2}.8^2+.....\right]=\dfrac{1}{8}+Integer$

$\therefore \left\{\dfrac{3^{200}}{8}\right\}=\left\{\dfrac{1}{8}+Integer\right\}=\dfrac{1}{8}$

19. Any point on the plane $=(5k-\alpha,2k+4,3k-4)$

D.r.'s of line $AD=(5k-\alpha,2k-1,3k-4-\alpha)$

$\because AD\perp BC\Rightarrow 5(5k-\alpha)+2(2k-1)+3(2k-4-\alpha)=0\Rightarrow 19k-4\alpha-7=0$ .....(i)

Area $\Delta ABC=21=\dfrac{1}{2}\times 2\sqrt{21}\times AD$

$\Rightarrow AD=\sqrt{21}\Rightarrow (5k-\alpha)^2+(2k-1)^2+(3k-4-a)^2=21$

$\Rightarrow 19k^2-8k\alpha+\alpha^2-14k+4\alpha=2$ .....(ii)

On solving equation (i) and (ii) , we get $\dfrac{x+\alpha}{5}=\dfrac{y-1}{2}=\dfrac{z+4}{3}$

20. $(a)\ B_1=A(adj\ A)=|A|.I_2$

$\therefore tr.(B_1)=2|A|=4$

$(b)\ B^2=A^2(adj(adjA))=A^2.|A|^{x-2}.A=A^3$

$\det.(B_2)=|A|^3=8$

$(c)\left|B_p\right|=|A|^{p+1}$

$\left|B_1^{-1}\right|+\left|B_2^{-1}\right|+\left|B_3^{-1}\right|+.....+\infty=\dfrac{1}{\left|A^2\right|}+\dfrac{1}{\left|A^3\right|}+\dfrac{1}{\left|A^4\right|}+.....\infty$

$=\dfrac{1}{4}+\dfrac{1}{8}+\dfrac{1}{16}+.....+\infty=\dfrac{\frac{1}{4}}{1-\frac{1}{2}}=\dfrac{1}{2}$

$(d)\ adj(adj\ B_2)=|B_2|^{n-2}\ B_2=B_2=A^3$

21. Define $a_n=f\left(\dfrac{1}{n}\right)(n=1,2,3,.....)$ then $a_1=f(1)=1$. For $x=-\dfrac{1}{K}\left(k\in N^+\right)$ Note that

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$\frac{x}{x-1}=\frac{\frac{-1}{k}}{-\frac{1}{k}-1}=\frac{1}{k-1}$ and $f(x)$ is an odd function

We have $f\left(\frac{1}{k+1}\right)=\frac{-1}{k}.f\left(-\frac{1}{k}\right)=\frac{1}{k}.f\left(\frac{1}{k}\right)$

and that means $\frac{a_{k+1}}{a_k}=\frac{1}{k}$. Therefore ,

$a_n=a_1\prod_{k=1}^{n-1}\frac{a_{k+1}}{a_k}=\prod_{k=1}^{n-1}\frac{1}{k}=\frac{1}{(n-1)!}$

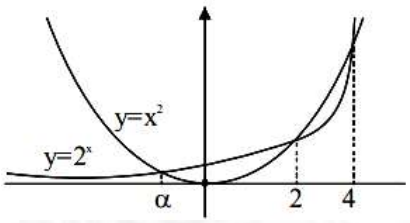
Then

$\sum_{i=1}^{50}a_i a_{101-i}=\sum_{i=1}^{50}\frac{1}{(i-1)!.(100-i)!}=\sum_{i=0}^{49}\frac{1}{i!.(99-i)!}$

$=\frac{1}{99!}.\sum_{i=0}^{49}{}^{99}C_i=\frac{1}{99!}.\sum_{i=0}^{49}\frac{1}{2}\left({}^{99}C_i+{}^{99}C_{99-i}\right)$

$=\frac{1}{99!}\times\frac{1}{2}\times 2^{99}=\frac{2^{98}}{99!}$

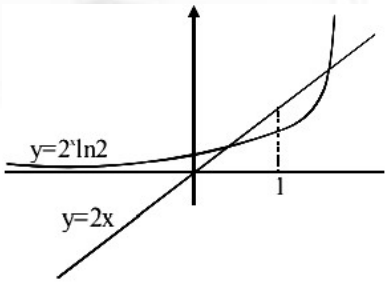
22. For the given conditions



Here point of intersections are 3,

$\therefore$ m = 43

For $f'(x)=e^x\ln 2-2x=0\Rightarrow 2^x\ln 2=2x$



Here point 2 of intersections are

$\therefore n=2\Rightarrow m+n=5$

23. Given variance of boys

$\sigma^2=2$ & $\bar{x}_b=12$ (average marks of boys) & variance of girls $\sigma_g^2=2$ & μ is average marks of girls

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Now, $\bar{x}_g = \mu = \dfrac{50\times 15 - 12\times 20}{30} = 17$

Combined variance

$$\sigma^2 = \frac{n_1\sigma_b^2 + n_2\sigma_g^2}{n_1+n_2} + \frac{n_1.n_2}{(n_1+n_2)^2}\left(\bar{x}_b - \bar{x}_g\right)^2$$

Where n_1 & n_2 are number of boys & girls

Given $n_1 = 20, n_2 = 30$

24. Let

$$I = \int_{-1}^{1} \cot^{-1}\left(\frac{1}{\sqrt{1-x^2}}\right).\left(\cot^{-1}\frac{x}{\sqrt{1-\left(x^2\right)^{|x|}}}\right)dx \quad .....(1)$$

$[P-5]$

$$I = \int_{-1}^{1} \cot^{-1}\left(\frac{1}{\sqrt{1-x^2}}\right).\left(\cot^{-1}\frac{-x}{\sqrt{1-\left(x^2\right)^{|x|}}}\right)dx \quad .....(2)$$

On Adding

$$2I = \int_{-1}^{1} \cot^{-1}\left(\frac{1}{\sqrt{1-x^2}}\right).\left\{\cot^{-1}\frac{x}{\sqrt{1-\left(x^2\right)^{|x|}}} + \pi - \cot^{-1}\frac{x}{\sqrt{1-\left(x^2\right)^{|x|}}}dx\right\}$$

$$2I = \int_{-1}^{1} \pi\cot^{-1}\left(\frac{1}{\sqrt{1-x^2}}\right)dx = \int_{-1}^{1} \pi\tan^{-1}\sqrt{1-x^2}\,dx$$

(As $\tan^{-1}\sqrt{1-x^2}$ is even function)

$$= 2\pi\int_0^1 \tan^{-1}\sqrt{1-x^2}\,dx \quad .....(3)$$

$$\therefore\ I = \pi\int_0^1 \underset{v}{1}\,\underbrace{\tan^{-1}\sqrt{1-x^2}}_{u}\,dx \quad .....(4)$$

Integrating by parts

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$$I=\pi\tan^{-1}\sqrt{1-x^2}.x\Big]_0^1-\int_0^1\frac{x}{(1+1-x^2)}\frac{(-x)}{\sqrt{1-x^2}}dx=0+\pi\int_0^1\frac{x^2}{(2-x^2)\sqrt{1-x^2}}dx$$

$Put\ \ x=\sin\theta\Rightarrow dx=\cos\theta$

$$I=\pi\int_0^{\pi/2}\frac{\sin^2\theta}{(2-\sin^2\theta)}d\theta=-\pi\int_0^{\pi/2}\frac{2-\sin^2\theta-2}{2-\sin^2\theta}d\theta\therefore I=2\pi\int_0^{\pi/2}\frac{d\theta}{2-\sin^2\theta}-\frac{\pi^2}{2}$$

$$I=2\pi\int_0^{\pi/2}\frac{\sec^2\theta d\theta}{2+2\tan^2\theta-\tan^2\theta}-\frac{\pi^2}{2}$$

$$=2\pi\int_0^{\pi/2}\frac{\sec^2\theta}{2+\tan^2\theta}-\frac{\pi^2}{2}\ put\ \tan\theta=t,I=2\pi\int_0^{\infty}\frac{dt}{2+t^2}-\frac{\pi^2}{2}$$

$$=2\pi\frac{1}{\sqrt{2}}\tan^{-1}\frac{t}{\sqrt{2}}\Big]_0^{\infty}-\frac{\pi^2}{2}=\frac{\pi^2}{\sqrt{2}}-\frac{\pi^2}{2}=\frac{\pi^2\left(\sqrt{2}-1\right)}{2}=\frac{\pi^2\left(\sqrt{a}-\sqrt{b}\right)}{\sqrt{c}}$$

$\Rightarrow a=2,b=1 and\ c=4\Rightarrow a+b+c=2+1+4=7$

25. Let us calculate when |R| = 0

Case – 1 ad = bc = 0

Now ad = 0

$\Rightarrow$ Total – (When none of a & d is 0)

$=8^2-7^2=15$ ways

Similarly bc = 0 $\Rightarrow$ 15 ways

$\therefore 15\times15$ ways of ad = bc = 0

Case – II ad = bc $\neq0$

either $a=d=b=c$

OR $a\neq d,b\neq d$ but $ad=bc$

${}^7C_1$ Ways

${}^7C_2\times2\times2=84$ Ways

Total 91 Ways

$\therefore|R|=0$ in 225 + 91 = 316 ways

$|R|\neq0$ in $8^4-316=3780$

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PHYSICS

26. $Q=\frac{a^4b^3}{c^2}$, $\frac{\Delta Q}{Q}=4\frac{\Delta a}{a}+3\frac{\Delta b}{b}+2\frac{\Delta c}{c}$ $=4+6+10=20\%$

27. Intensity $I=4I_o\cos^2\frac{\phi}{2}$, In YDSE energy conservation is followed

28.

$$a=\frac{m_2-m_1}{m_1+m_2}g=\frac{\lambda L-\lambda l-\lambda l}{\lambda L}g,\ a=\frac{L-2l}{L}g \quad \frac{g}{3}=\frac{L-2l}{L}g \quad l=\frac{L}{3}$$

29. **Conceptual** $I=\frac{ml^2}{3}$

30. Since $T^2=Kr^3 \Rightarrow \frac{2\Delta T}{T}=3\frac{\Delta r}{r}\Rightarrow \frac{\Delta T}{T}=\frac{3}{2}\frac{\Delta r}{r}$

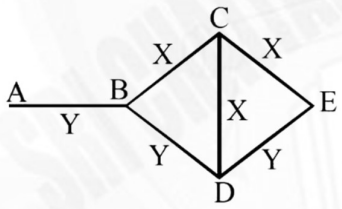
31. As length is same

So $\left(\frac{F}{A}\right)_{compressive}=\left(\frac{F}{A}\right)_{Thermal}$

$$\frac{F}{\pi r^2}=Y\alpha T$$

Linear coefficient of expansion $\alpha=\frac{F}{YT\pi r^2}$

32.



No heat flowing through the rod CD
Resistance of X is R
Resistance of Y is R/2

$$R_{eq}=R_{AB}+R_{BE},R_{BE}=\frac{2R\times R}{2R+R}=\frac{2R}{3}$$

$$R_{eq}=\frac{R}{2}+\frac{2R}{3}=\frac{7R}{6}=\frac{7l}{6KA}$$

33. $v_{rms}=\sqrt{\frac{3RT}{M}},v_p=\sqrt{\frac{2RT}{M}}$

34. $Kx=\omega t$ $,wavevelocity=\frac{x}{t}=\frac{\omega}{K}$

35. NAND and NOR gates are called universal gates

36. $Q=\frac{q}{2}$

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37. The charge = Q = 5.4μF x 10 V = 54 μF
The charge in branch PQ, Q_{PQ} = 24 μF x 10 V = 24 μC
Since 4μF is in series with 6μF (=1μF || 5μF), So $\theta_{4\mu F} = 24\mu C$

38. $I = \frac{V}{R} = \frac{5}{50} = \frac{1}{10},\ V_{10\Omega} = I \times R = 2V,\ Q = C \times V_{10\Omega} = 16\mu C$

39. $r = \frac{mv}{Bq} = \frac{\sqrt{2mK}}{Bq}$

40. Application of electromagnetic waves

41. $\frac{F}{l} = \frac{\mu_o i_1 i_2}{2\pi d}$

42. $\mu_r = 1 + \chi$

43. If internal forces acting, centre of mass remains at rest.

44. Since it is a balanced wheatstone bridge, its equivalent resistance $\varepsilon = Blv = 5 \times 10^{-4} V$
So total resistance $R = \frac{4}{3} + 2.7 \approx 4\Omega, i = \frac{\varepsilon}{R} \approx 125\mu A$

45. Statement 1 : Inside conductor
Statement 2 : Dipole

46. $KE = PE,\ \frac{1}{2}m\omega^2\left(A^2 - x^2\right) = \frac{1}{2}m\omega^2\left(x^2\right),\ x = \frac{A}{\sqrt{2}}$

47. **Power factor** $\cos\phi = \frac{R}{Z} = \frac{R}{\sqrt{R^2 + X_L^2}} = \frac{1}{\sqrt{10}}$

New **Power factor** $\cos\phi = \frac{R}{Z} = \frac{R}{\sqrt{R^2 + (X_L - X_C)^2}} = \frac{R}{\sqrt{2}R}$

48. $\frac{1}{f} = \frac{1}{v} - \frac{1}{u}, \frac{1}{v} - \frac{1}{-12} = \frac{1}{10},\quad i.e\ v = 60cm$
In the absence of convex mirror , convex lens will form the image at a distance of 60 cm behind the lens. Since the mirror is at a distance of 10cm from the lens, the image will be at a distance of 50 cm from the mirror. i.e. R =50 cm , then f =25 cm

49. $I = K\left(\cos^2\frac{\phi}{2}\right),\ \phi = \frac{2\pi}{\lambda}(\Delta x)$

50. $\frac{1}{\lambda} = R\left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right)$
First transition line of Lyman series

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CHEMISTRY

51. A → (IV) Combination
B → (I) Decomposition
C → (II) Displacement
D → (IV) Disproportionation

52. The group – 15 elements, pentavalent oxides E_2O_5, are more acidic than trivalent oxides E_2O_3, of the same elements.
The trivalent oxides acidic strength decreases as metallic strength increases down the group.

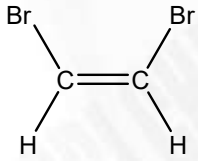
53. $2Cu^+ \rightarrow Cu^{2+} + Cu$
The stability of $Cu^{2+}(aq)$ is more than $Cu^+(aq)$ due to the much more negative hydration enthalpy of $Cu^{2+}(aq)$ than $Cu^+(aq)$

54. The ligand, $EDTA^{4-}$ is Hexadentate ligand. Hence for complex, $\left[Mg(EDTA)\right]^{2-}$, coordination environment is octahedral.

55. P^{-3}, S^{-2}, Cl^{-1} **and** Ca^{+2}

56. NO_2 **and** H_3PO_4

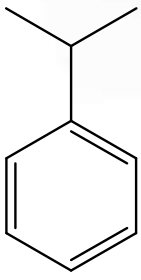
57. $NH_2 - \underset{\underset{S}{\|}}{C} - NH - NH_2$

58. 

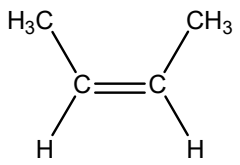
59. Stability of carbocation is depends on no. of resonating structures.
The more resonating structures, the more will be the stability or carbocations.

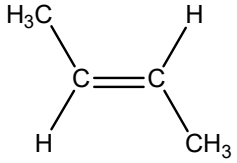
60. Aromatic amines < Ammonia < Aliphatic amines

61. Fredel crafts alkylation

62.  acetic acid (CH_3COOH)

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63. (A) (B)

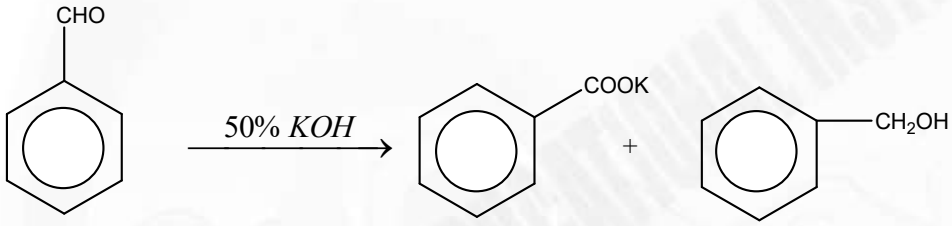
64. From the diagram it can be interpreted that,
A → weak electrolyte
B → strong electrolyte

65. Tyrosine is an essential amino acid.

66. Proline

67. NCERT concept

68. Potassium benzoate and benzyl alcohol will formed as products
When benzaldehyde treated with 50 % KOH and the reaction is known as Cannizzaro reaction.



69.

Group 14	Electromagnetic value
Carbon	2.5
Silicon	1.8
Germanium	1.8
Tin	1.8

70. O_2 can form $2p\pi - 2p\pi$ multiple bond with itself due to its small size while, S cannot form multiple bond with itself as $3p\pi - 3p\pi$ bond are not stable due to bigger size of S, but S can form multiple bond with small size atom like N and C.
For e.g: $S=C=S$

$S=C=N \leftrightarrow S^{-}-C\equiv N$

71. $N_2(g)+3H_2(g) \rightleftharpoons 2NH_3(g), K_1 = 4\times10^5$ .....(i)

$N_2(g)+O_2(g) \rightleftharpoons 2NO(g), K_2 = 1.6\times10^{12}$ .....(ii)

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$$H_2(g)+\frac{1}{2}O_2(g)\rightleftharpoons H_2O(g), K_3=1.0\times10^{-13} \quad .....(iii)$$

$(ii)+3\times(iii)-(i)$

$$2NH_{3(g)}+\frac{5}{2}O_2(g)\rightleftharpoons 2NO(g)+3H_2O(g)$$

$$k_{eq}=\frac{k_2\times k_3^3}{k_1}=\frac{1.6\times10^{12}\times\left(10^{-13}\right)^3}{4\times10^5}=\frac{1.6}{4}\times10^{-32}=4\times10^{-33}$$

72. In 0.001 KOH solution, the concentration of $\left[OH^-\right]$ is $10^{-3}\,M$

As $pOH=-\log_{10}[OH^-]=-\log_{10}10^{-3}\Rightarrow pOH=3$

Also $pH+POH=14$

$pH-14=POH$

$pH=14-3$

$pH=11$

73. The combustion reactions of propane and butane are :

$C_3H_8+5O_2\rightarrow3CO_2+4H_2O$

$2C_2H_6+7O_2\rightarrow4CO_2+6H_2O$

$\therefore 2\,mol\,C_2H_6$ is given

$\therefore 7\,mol\,O_2$ is needed for combustion of ethane.

Total mole of $O_2=5+7=12$

74. $r\alpha\frac{n^2}{Z}$

Given : $r_H=0.6\,\overset{o}{A}$

n = 3 (For 3rd orbit)

z = 3 (for $Li^{+2}\,ion$)

$$r_{Li^{+2}}=r_H\times\frac{n^2}{Z}$$

$$r_{Li^{+2}}=0.6\times\frac{3^2}{3}=1.8\,\overset{o}{A}$$

$$r_{Li^{+2}}=180\,pm$$

$\Delta U=Q+W$

75. $\Delta U=20-15$

$\Delta U=5$

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